

• Medium → #5 → Longest Palindromic Substring

▲ Problem Statement → Given a string "s", find & return the longest substring that is a palindrome.

▲ Main Solution → Treat every position as the center of a possible palindrome & expand left & right while the characters match.

like check two cases :- • Odd Length → One character is the center

• Even Length → The center is between two characters.

▲ Brute Force → Check every possible substring & see if it is a palindrome. Keep the longest one.

▲ Algorithm → 1 → Store "start = 0" & "maxLength = 1"

2 → For every index "i" :- • Check an odd palindrome using "left = i" & "right = i".

• Check an even palindrome using "left = i" & "right = i + 1"

3] Expand left & right while: • Indices are inside the string

• Characters are equal.

4] If the palindrome is longer than "maxLength", update "start" & "maxLength".

5] Return the substring using "start" & "maxLength".

▲ Important Points → • A palindrome reads the same forward & backward.

• Every index can be the center of an odd-length palindrome.

• Every gap between two indices can be the center of an even-length palindrome.

• Expand using "left--" & "right++".

• Save the starting index & length of the longest palindrome.

▲ Complexity → Brute Force → $O(n^3)$ & $O(1)$
Main Solution → $O(n^2)$ & $O(1)$